

MAINAK CHANDRA

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PROFESSIONAL SUMMARY

Engineering professional with 7 years of experience across test automation, production reliability, cloud/Linux systems, and release validation. Hands-on with Java, Python, Selenium WebDriver, TestNG, SQL, APIs, and debugging. At PwC, validated 350,000+ learner accounts and built AI automation that reduced manual reporting effort by up to 95%. Built production-grade automation and software projects spanning semantic search, API integrations, CI/CD, and web engineering.

TECHNICAL SKILLS

Programming Languages: Java, Python, TypeScript, SQL (T-SQL)

Test Automation: Selenium WebDriver, TestNG, pytest, Page Object Model (POM), Apache POI, Data-Driven Testing, XPath, Assertions, Explicit Waits, Test Listeners

API & Backend: REST APIs, Next.js API Routes, JSON, API Integration, Data Validation, Error Handling, Data Normalization, Caching, Fallback Strategies

Machine Learning & Semantic Search: Sentence Transformers, Text Embeddings, Semantic Search, Cosine Similarity, Ranking & Scoring Algorithms, Resume Parsing

Databases: MySQL, Microsoft SQL Server

Cloud, Systems & Reliability: AWS EC2, IAM, CloudWatch, S3, VPC, Linux, Debugging, Root Cause Analysis (RCA), Production Monitoring, Release Validation

Web Engineering: Next.js, React

AI Platforms: Microsoft Copilot Studio

Build, CI/CD & Engineering Practices: Git, GitHub Actions, Gradle, OOP, Agile/Scrum, STLC

PROFESSIONAL EXPERIENCE

PricewaterhouseCoopers (PwC) Services LLP

Senior Associate | July 2021 – Present

- Led pre-launch validation, risk assessment, and go-live assurance for global LMS implementations and integrations spanning HRMS, Active Directory, and third-party content providers; coordinated engineering, integration, and business stakeholders across release cycles.
- Validated and reconciled 350,000+ learner accounts during a large-scale LMS migration, applying completion-precedence rules and migrating exclusive training records to preserve learning-history continuity.
- Investigated complex multi-platform production issues through structured RCA and SQL-based troubleshooting; drove cross-functional resolution and recurrence-reduction actions while maintaining SLA and audit requirements.
- Built AI agents in Microsoft Copilot Studio to automate ServiceNow ticket trend analysis, reducing manual reporting effort by up to 95% and producing KPI and trend outputs for monthly and quarterly reviews.
- Standardized support workflows, SOPs, runbooks, and operational procedures to improve repeatability across incident, change, problem, release, and go-live activities.

Tata Consultancy Services (TCS)

Systems Engineer | June 2019 - July 2021

- Managed Linux systems on AWS EC2 across RHEL, CentOS, and Ubuntu; monitored CPU, memory, disk, CloudWatch metrics, and Linux system logs to diagnose reliability and performance issues.
- Automated patch management and infrastructure provisioning using AWS Systems Manager Patch Manager, Ansible, Terraform, and CloudFormation.
- Configured IAM, security groups, firewalls, VPC/networking, storage, and backup/disaster-recovery controls using AWS Backup, EBS snapshots, and Route 53 failover.

PROJECTS

Semantic Job Matcher | [GitHub](#)

- Built a multi-source Python job aggregation pipeline integrating public ATS APIs, official company career portals, and Gmail job alerts into a standardized job schema.
- Engineered fault-isolated source processing and cross-source deduplication, preventing individual source failures from disrupting the pipeline and eliminating duplicate job listings.
- Implemented semantic job ranking using Sentence Transformers and cosine similarity, with weighted scoring across job title, skills, and experience requirements.
- Automated scheduled job-matching runs and batched Gmail notifications with persistent notification state; containerized the application using Docker and added pytest-based automated test coverage.

3D Dynamic Portfolio Platform | [Live](#) | [GitHub](#)

- Built and deployed a production-grade portfolio platform using Next.js 14, TypeScript, React, React Three Fiber/Three.js, Framer Motion, and Vercel, with responsive UI and interactive 3D visualization.
- Implemented SEO metadata, analytics, and automated Vercel redeployment, improving production readiness, observability, and deployment efficiency.
- Designed server-side API routes to synchronize GitHub activity, LeetCode statistics, and Crio projects, with caching, validation, deduplication, and graceful fallback handling for unreliable upstream services.
- Developed a resume-grounded AI assistant using local retrieval with optional OpenAI-backed responses, automatically falling back to verified local context when external provider configuration or API calls are unavailable.

QKart QA Automation | [GitHub](#)

- Built an end-to-end Java/Selenium/TestNG framework covering registration, login, product search, cart, and checkout; implemented reusable POM components, dynamic XPath, explicit waits, Apache POI data-driven testing, parameterization, grouping, assertions, and listener-based screenshots for failure diagnosis.

EDUCATION

B. Tech, Electronics and Communication Engineering | National Institute of Technology, Arunachal Pradesh | 2015-2019 | CGPA: 7.32